Transforming Production Planning Through Data Intelligence

Project Plan document

Egemen Alkan
Student Bachelor Applied Computer Science

2025 - 2026

Table of Contents

1. INTRODUCTION & BUSINESS CASE (THE 'WHY') 3

2. PROJECT CONTEXT & STAKEHOLDERS (THE 'WHO') 3

2.1. Project Team 3

2.2. Primary End-Users 3

2.3. User Needs & Pains 3

3. PROPOSED SOLUTION & CONCEPT (THE 'WHAT') 4

3.1. Primary Deliverable: The Enhanced Weekly Planning Report 4

3.2. Secondary Deliverable: AI Planning Assistant (Research & Prototyping) 5

3.3. Supporting Documentation 6

3.3.1. The Weekly Planning Report 6

3.3.2. The AI Planning Assistant Research 7

4. METHODOLOGY & TOOLS 7

4.1. Phase 1: Discovery, Alignment, Planning (Weeks 1-2) 7

4.2. Phase 2: Power BI Report Development, Testing & Automation (Weeks 3-5) 7

4.3. Phase 3: AI Assistant Research & Development (Weeks 6-11) & Final Documentation & Handover (Week 12) 8

5. EXPECTED ADDED VALUE 9

1. Introduction & Business Case (The 'Why')

The current process for preparing for the weekly planning meeting is a time-consuming, manual effort that relies on gathering data from multiple sources. This often leads to a significant portion of the meeting being spent on reporting facts rather than strategic analysis. As a result, the planning process can be reactive, focusing on solving immediate issues rather than proactively identifying future risks and opportunities.

This project aims to address these challenges directly. The primary objective is to transform the planning process by trying to implement planning standards that are missing into reports and developing tools that make it more efficient, data-driven, and proactive. By automating the core reporting tasks and providing intelligent support, the Planning Team can shift its focus from data compilation to strategic decision-making, ensuring the production plan is robust, efficient, and aligned with business goals.

2. Project Context & Stakeholders (The 'Who')

2.1. Project Team

Intern: Egemen Alkan
Work Placement Mentor: Wietse Beterams

2.2. Primary End-Users

The primary users of this project's deliverables are the members of the Planning Team.

2.3. User Needs & Pains

Based on initial analysis, the Planning Team requires a solution that addresses the following needs:

- A consolidated, reliable 6-week forward view of the production plan.
- Flagging of production orders that violate established standard planning rules.
- Clear visibility into setup group sequencing and alerts for excessive changeovers.
- Compliance checks for Minimum/Maximum Order Quantities (MOQ/MAXQ) and preferred production lines.
- A faster method to identify and prioritize orders based on operational risk.

3. Proposed Solution & Concept (The 'What')

This project is strategically divided into a primary, high-priority deliverable (The Power BI Report) and a secondary, exploratory research objective (The AI Assistant). This phased approach ensures that the most critical business needs are met first, aligning with the core objectives of the internship contract.

3.1. Primary Deliverable: The Enhanced Weekly Planning Report

- **The 'Why' (Business Need):** The Planning Team's current process for preparing for the weekly operations meeting is a significant operational bottleneck. It involves manually gathering data from different sources, consolidating it, and formatting it for presentation. This process is not only time-consuming but also introduces a risk of human error and data inconsistency. Consequently, a substantial portion of the meeting is often dedicated to validating the data itself, rather than performing strategic analysis. This reactive cycle limits the team's ability to look ahead and make proactive, data-driven decisions.

The primary goal of this deliverable is to fundamentally transform this workflow. By creating an automated, centralized, and reliable "single source of truth," we eliminate the administrative burden of meeting preparation. This will empower the Planning Team by shifting their focus from data compilation to strategic action, transforming the weekly meeting from a historical review into a forward-looking, decision-making forum.

- **The 'What' (Key Features):** The project will enhance the existing Power BI report by implementing a comprehensive "Deviation Tracker." This is not merely a collection of charts, but a sophisticated analytical engine designed to translate the 10 "golden" planning rules from abstract concepts into clear, tangible, and visual alerts. This allows planners to see, at a glance, the health of the production plan.

The key features to be implemented are:

- **Plan vs. Demand Analysis:** This feature provides a strategic overview of the plan's alignment with commercial goals. By visualizing the production plan against the latest demand forecast, it helps the team manage inventory levels effectively and prevent costly overproduction or customer-impacting stock shortages.

- **Order Quantity and Duplicate Item Checks:** This feature enforces supply chain discipline and highlights potential data errors or inefficiencies. Automated flags will identify orders that violate Minimum Order Quantity (MOQ) rules, which can cause issues with suppliers. A statistical model will also flag potential Maximum Order Quantity (MAXQ) violations by identifying orders with unusually large quantities, which could indicate data entry errors or planning anomalies that might strain production capacity. Furthermore, a Duplicate Item Check identifies orders that may be inefficiently planned. An order is flagged as a "Potential Inefficient Duplicate" if another order for the same product has been planned within the previous 28 days, but only if the order size is below its calculated statistical MAXQ. This prevents flagging large, intentionally separate orders while highlighting smaller, frequent orders that could potentially be consolidated.

- **Production Version (PV1) Adherence:** This tracker focuses on cost and quality control. By ensuring orders are planned on their standard, most efficient production line, it helps maintain process consistency and prevent the use of less optimal, potentially more expensive production methods.

- **Frequency and Changeover Monitoring:** This provides two levels of efficiency monitoring, focusing on both short-term changeover discipline and long-term planning cadence. The Weekly Changeover per Resource monitor tracks the number of "true" setup group changes on each production line per week. The calculation is sophisticated: it understands that a changeover for "Tube" lines (where the setup group code starts with "T") only occurs if the first 5 characters of the code change, while for all other "Vial/LMU" lines, a changeover is triggered by a change in the first 8 characters. This provides a more accurate measure of costly changeovers than a simple count. The Planning Frequency monitor checks if setup groups are being planned according to the "campaigning" efficiency rule (i.e., not run too frequently). As a "sequence" or required run cycle is not explicitly defined in the data model, this analysis uses a derived metric. It calculates the Median time between all historical runs for each setup group to determine its "Typical Sequence." It then flags a violation if the shortest gap between runs in the current planning period is shorter than that setup group's own typical sequence. While this is an effective data-driven approach, it is an inferred rule and should be validated with planners.

- **Automated Refresh & Distribution:** This is the final step for the administrative tasks. The report will be configured for a daily scheduled refresh, ensuring the most current and accurate information is always in the hands of the decision-makers before their meeting.

3.2. Secondary Deliverable: AI Planning Assistant (Research & Prototyping)

- **The 'Why' (Business Need):** While the enhanced Power BI report is excellent at showing *what* is deviating from the plan, it doesn't always tell a planner *what is most important* among hundreds of potential signals. Planners, especially those who are newer to the role, also may not immediately know the correct procedure to follow when a problem is identified. An AI assistant has the potential to solve these two challenges: prioritization and guided action. The purpose of this workstream is to research the feasibility of an AI "Co-Pilot" that could proactively surface the most critical risks and provide planners with knowledge assistance from official company documentation.

- **The 'What':** Due to the time it takes to finish the first deliverable and the complexities of securing intern access to enterprise-level AI platforms like Microsoft Copilot Studio, this workstream is formally defined as a research and feasibility study. This is a standard practice for corporate AI initiatives, designed to de-risk the project and ensure a strong business case before committing to full-scale development.

  - **Primary Outcome:** The main deliverable will be a recommendation document and presentation. This document will detail the findings on feasibility, a proposed technical architecture, an analysis of the required data, a summary of the potential business value (ROI), and a final recommendation on whether the company should proceed with development.

  - **Secondary (Stretch) Goal:** If access to the necessary tools becomes available and time permits after the completion of the primary Power BI deliverable, a working prototype (Proof-of-Concept) will be developed. This prototype will not be a full solution but will aim to demonstrate the "art of the possible" to stakeholders by showcasing one or two key functionalities.

- **Areas of Investigation:** The research will focus on the technical implementation of two core capabilities:

  - **A Proactive Risk-Scoring Engine:** This investigation goes beyond the simple "red/green" flags in the Power BI report. The goal is to design a weighted scoring system that can rank production orders by overall risk level. The research will involve interviewing senior planners to understand how *they* weigh different risks (e.g., "is a material shortage more critical than an expiration risk?") and then translating that human expertise into a logical model.

  - **A Q&A Knowledge Base:** This investigation will explore using a technique called Retrieval-Augmented Generation (RAG). The research will focus on how a Copilot could be connected to the company's SharePoint site. When a planner asks a question, the AI would first *retrieve* relevant passages from the official SOP documents and then use that information to *generate* a summarized, conversational answer. This ensures that the advice given by the AI is always based on official, up-to-date documentation.

3.3. Supporting Documentation

A complete handover package will be provided to ensure the solutions are maintainable, and the research is actionable.

3.3.1. The Weekly Planning Report

While the report is designed to be user-friendly, the complex business logic built into the DAX measures requires a technical reference for future maintenance and validation.

- **Business Rules & DAX Logic Guide:** This is the core technical document. It will detail how each of the "10 golden planning rules" was translated into a specific DAX measure or calculated column. This is essential for understanding how violations are calculated and for making future adjustments.

- **Data Model Overview:** A concise summary of the key tables, relationships, and any specific modelling decisions made to enable the calculations.

3.3.2. The AI Planning Assistant Research

This is the primary written deliverable and will contain all findings from the research workstream.

- **Feasibility Study & Formal Recommendation:** The main outcome of the research, this document will detail the business case, potential architecture, estimated value, and a formal go/no-go recommendation for a full-scale AI assistant project.

- **Prototype Documentation (Conditional):** If a proof-of-concept prototype is developed, this documentation will be included. It will contain:
  - **AI Logic & Workflow Diagram:** A clear map of the decision-making process built into the AI agent.
  - **Technical Setup Guide:** Instructions for configuring and running the prototype, including any necessary platform or data connection details.

4. Methodology & Tools

This project will be executed in four distinct phases, with a clear focus on completing the primary Power BI deliverable first, followed by the research and prototyping of the AI assistant.

4.1. Phase 1: Discovery, Alignment, Planning (Weeks 1-2)

Weeks 1-2 (Research & Planning): The initial two weeks were dedicated to understanding the project context. This involved initial meetings, a deep dive into the existing Power BI report and its underlying data model, and a preliminary analysis of the 10 "golden" planning rules and also research into Copilot AI agents and how to achieve the assignment goals.

4.2. Phase 2: Power BI Report Development, Testing & Automation (Weeks 3-5)

This three-week phase is now dedicated to the end-to-end development and finalization of the primary internship deliverable.

- **Week 3: Core Logic & Foundational Visuals**
  - **Primary Focus:** To finalize the core DAX logic and build the main report dashboards.
  - **Key Tasks:**
    - Develop and test the majority of the DAX calculated columns and measures required to track the planning rules.
    - Construct the "Plan Meeting" dashboard and the first drafts of the detailed deep-dive pages.
    - Finalize the remaining complex DAX calculations for the last 1-2 rules that are still in progress.

    - **Weekly Deliverable:** A functional report with most core visuals and logic in place, ready for a detailed review.

- **Week 4: Final Visuals, Mentor Review & Feedback**
    - **Primary Focus:** To complete all visuals and conduct a final review session with my work mentor.
    - **Key Tasks:**
        - Build the remaining advanced visuals.
        - Walk through the completed report and gather final feedback with the workplace mentor.
        - Document all feedback and create a final list of required revisions.
    - **Weekly Deliverable:** A feature-complete report, fully reviewed by my work mentor, with a clear list of final revisions.

- **Week 5: Revisions, Automation & Publication**
    - **Primary Focus:** To finalize the report and implement the full automation workflow.
    - **Key Tasks:**
        - Implement all revisions based on the mentor's feedback.
        - Publish the final, approved report to the Power BI Service.
        - Configure the Scheduled Refresh and ensure the Data Gateway is working correctly.
        - Build, test, and activate the Power Automate flow for distribution.
    - **Weekly Deliverable:** The final published Power BI report is live and automatically refreshed.

4.3. Phase 3: AI Assistant Research & Development (Weeks 6-11) & Final Documentation & Handover (Week 12)

- **Weeks 6-7: AI Feasibility & Logic Design**
    - **Primary Focus:** To conduct a thorough feasibility study and design the core logic for the AI assistant.
    - **Key Tasks:** Finalize confirmation of tool access, try to map out the Risk-Scoring Engine logic, and research the end-to-end architecture.
    - **Weekly Deliverable:** A formal recommendation document.

- **Weeks 8-10 (Conditional): AI Backend & Core Development**
    - **Primary Focus:** If prototyping is approved, the focus will be on building the data connection and core conversational logic.
    - **Key Tasks:** Build and test Power Automate flows, design Adaptive Cards, and construct foundational topics in Copilot Studio.
    - **Weekly Deliverable:** Tested backend flows and a basic Copilot structure.

- **Week 11 (Conditional): AI Frontend, Testing & Refinement**
    - **Primary Focus:** Building the user experience and gathering feedback on the prototype.
    - **Key Tasks:** Refine trigger phrases, connect to the SharePoint knowledge base, and test the AI assistant in Microsoft Teams.

- **Weekly Deliverable (Stretch Goal):** A working prototype of the AI assistant.

- **Alternative Plan (If Prototyping is Not Feasible):** If tool access is unavailable, Weeks 8-11 will be dedicated to the 'Deep Dive' Research Phase to do more detailed research on various different ways to achieve the AI Assistant including ways other than using Copilot Studio.

5. Expected Added Value

The successful delivery of this project is expected to provide the following benefits:

- **Reduce Meeting Preparation Time:** Cuts down the hours spent on compiling data.
- **Enhance Decision-Making:** Provides clear, fact-based insights to support strategic discussions.
- **Increase Planning Efficiency:** Helps planners quickly identify and address deviations from standard procedures.
- **Improve Proactive Risk Management:** Shifts focus from solving past problems to anticipating future challenges.